

ArchiHotel — Reservation Handling & Check-In

Team C2F1G4 — Katrin Schwab · Soyeong Jeong · Fiona Aurelia · Angel Choi · Demir Eroglu

Technical University of Munich — School of Computation, Information and Technology

Enterprise Architecture Management and Reference Models — Case Study Final Presentation

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Our EAM Team



One layer per person — merged into one model



Katrin Schwab

Abstract EA view · model
QA & integration



Soyeong Jeong

Business layer —
Reservation & Check-In



Fiona Aurelia

Application layer — HMS
internals



Angel Choi

Cross-layer alignment &
consolidation

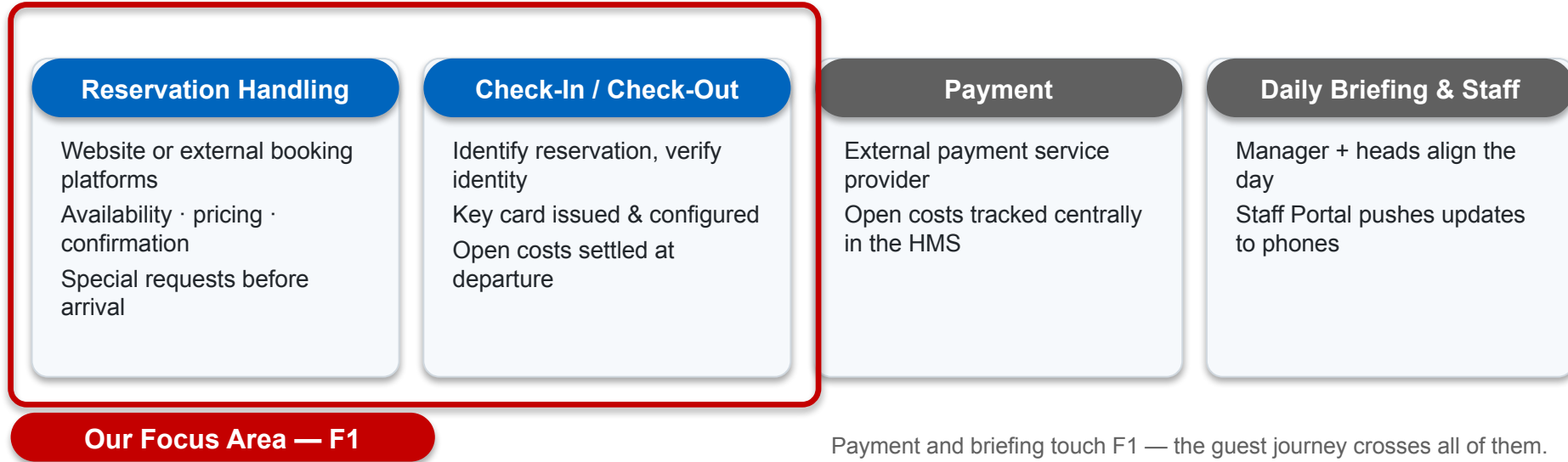


Demir Eroglu

Technology layer — key-
card stack

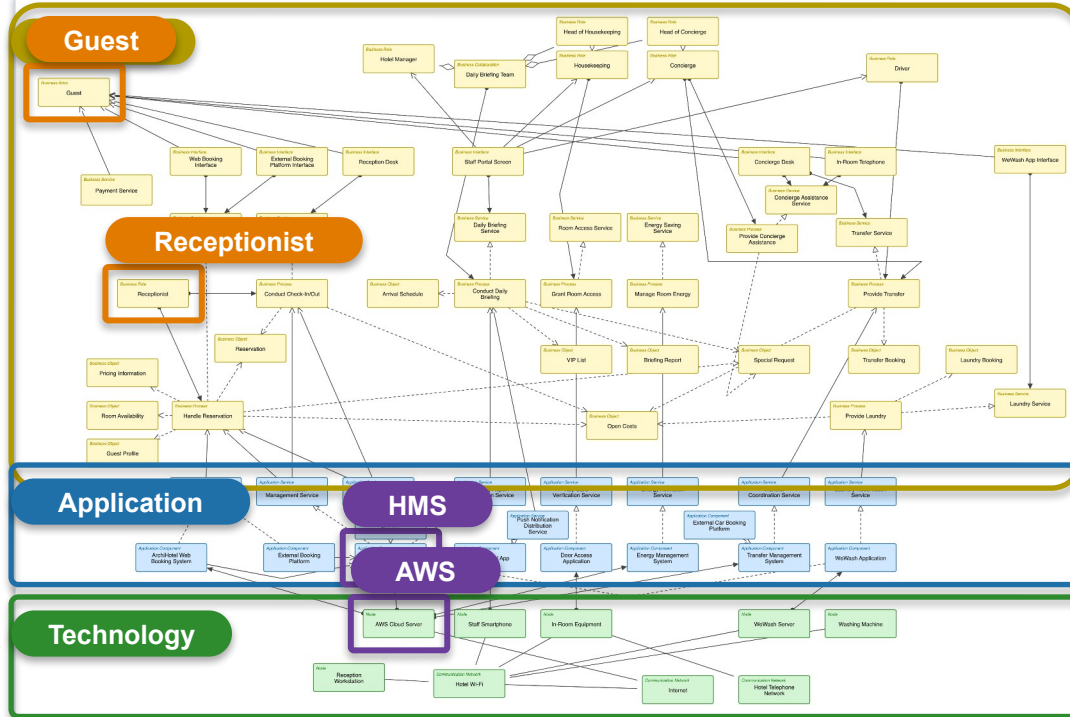
The Case: ArchiHotel's Services

A luxury hotel where “luxury feels effortless” — run on one Hotel Management System



The Abstract EA Model — the Map

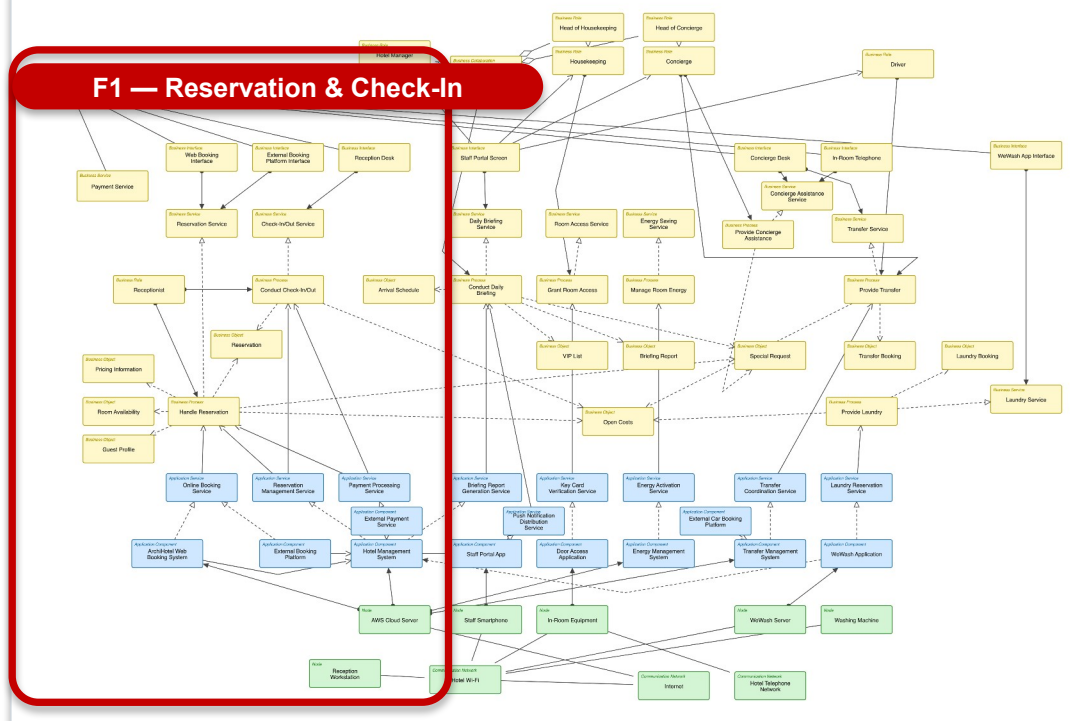
Don't read the boxes — watch the structure. 72 elements · 98 relationships on this view



- **Three ArchiMate layers**
Business → Application → Technology
- **Key stakeholders**
Guest and Receptionist drive F1
- **The convergence point**
HMS on the AWS Cloud Server

Focus Area F1 — Where We Zoom In

Handle Reservation → Conduct Check-In/Out — the guest's pre-arrival to arrival journey



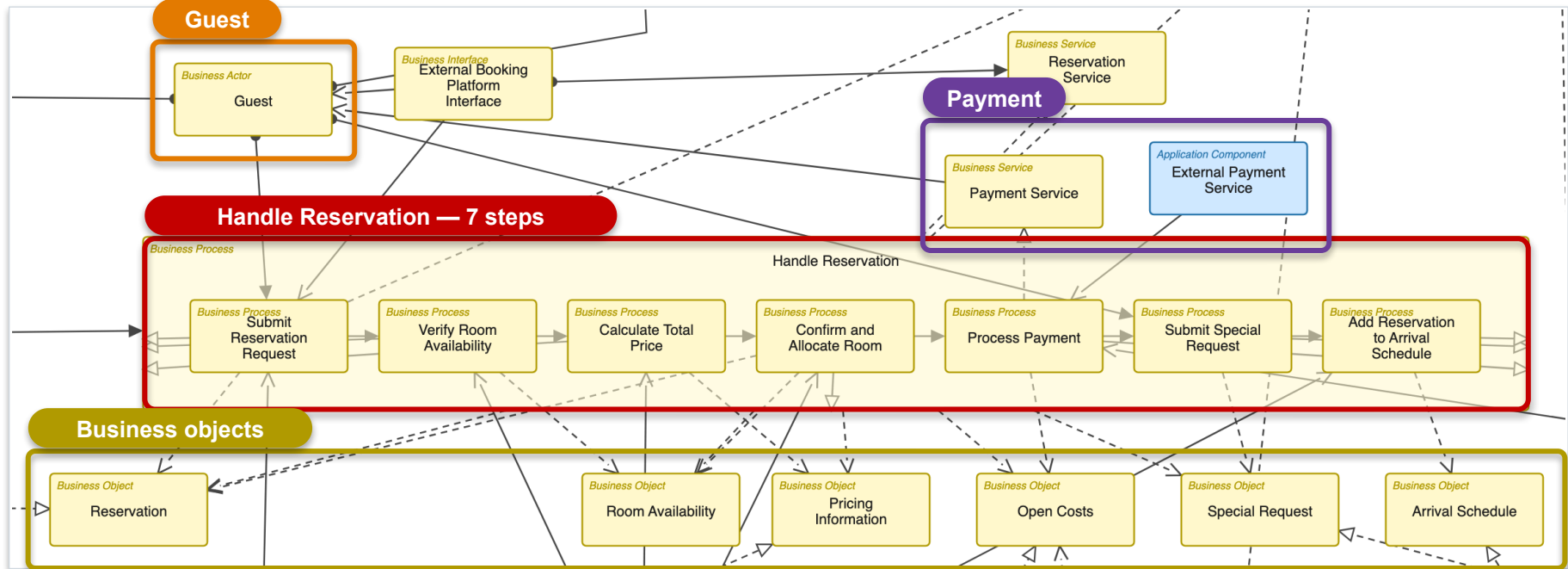
1 Business layer
Who does what — processes & objects

2 Application layer
Which systems support it — the HMS

3 Technology layer
What it runs on — down to the door

F1 Business Layer — Reservation

One abstract process becomes seven concrete steps



F1 Business Layer — Check-In

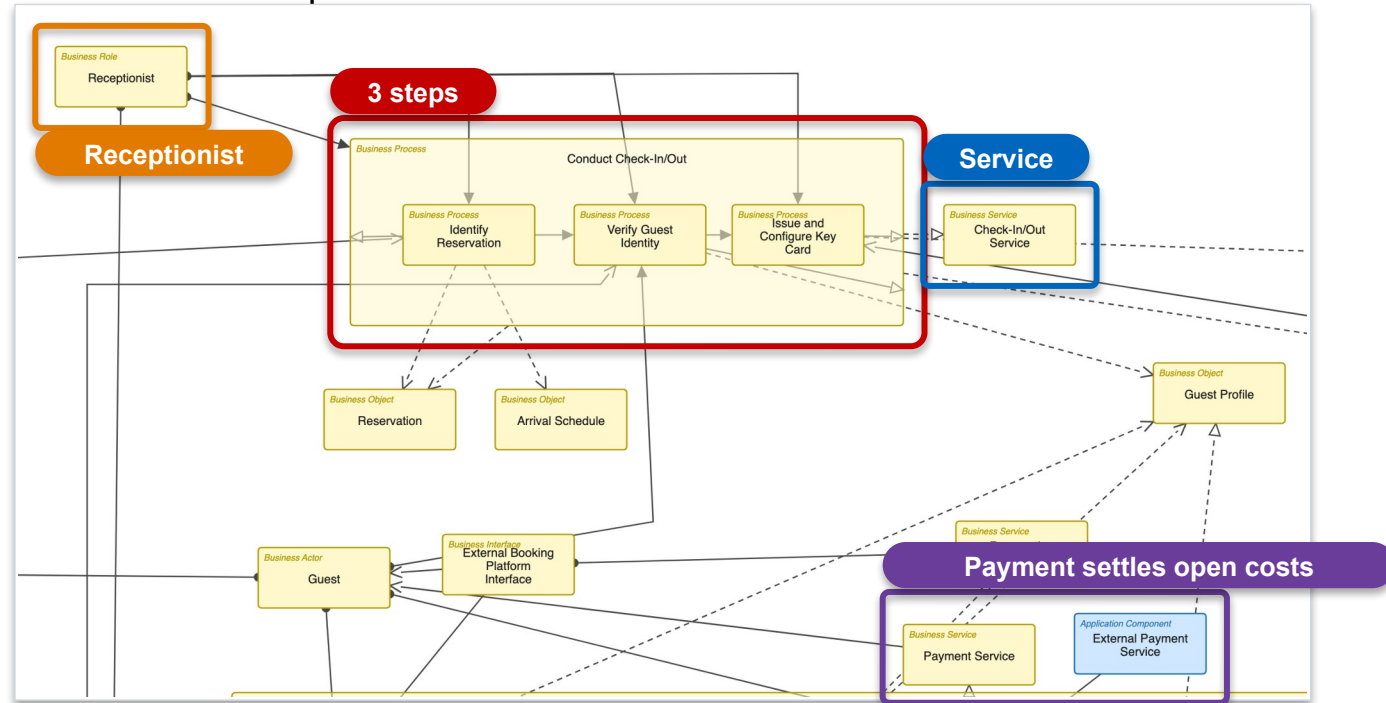
The Receptionist turns a reservation into an open door

The Receptionist is assigned to the whole process.

Identify Reservation → Verify Guest Identity → Issue and Configure Key Card.

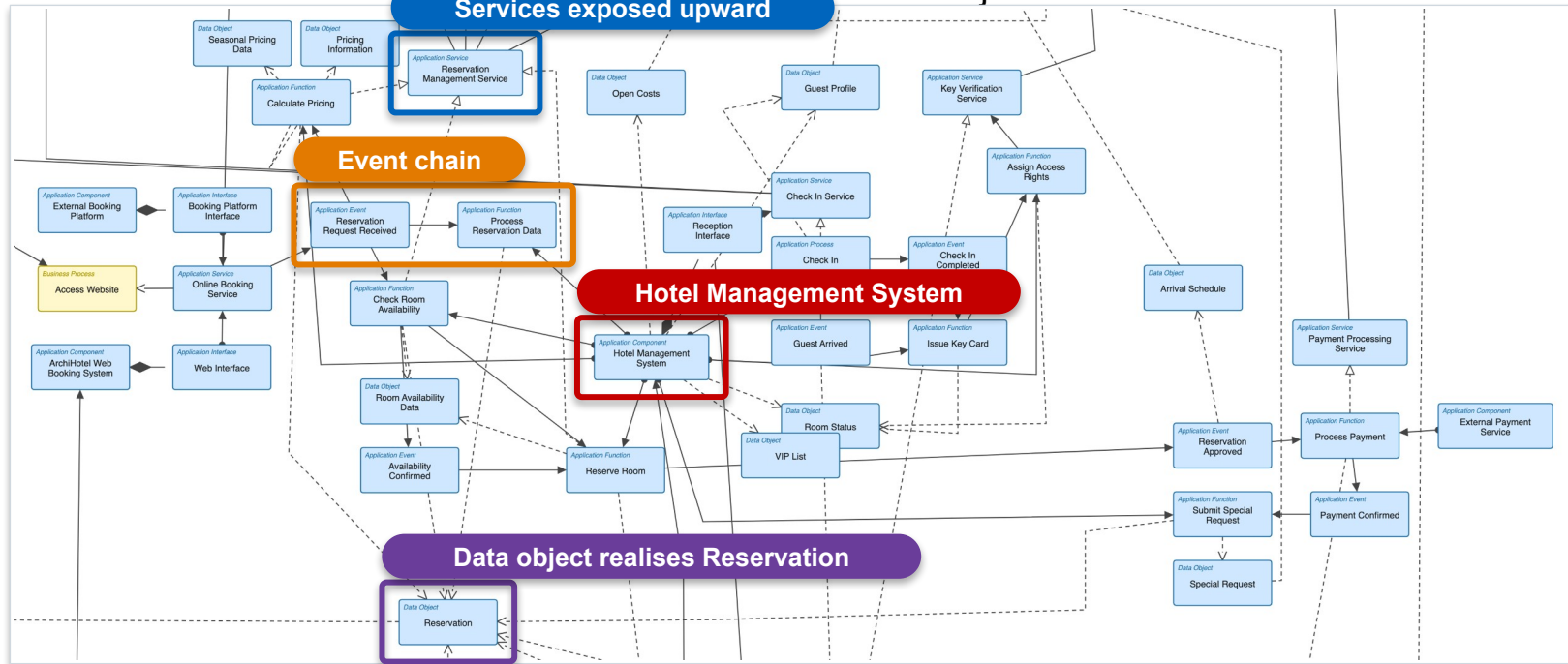
Realises the Check-In/Out Service the guest sees.

Open costs settle via the Payment Service.



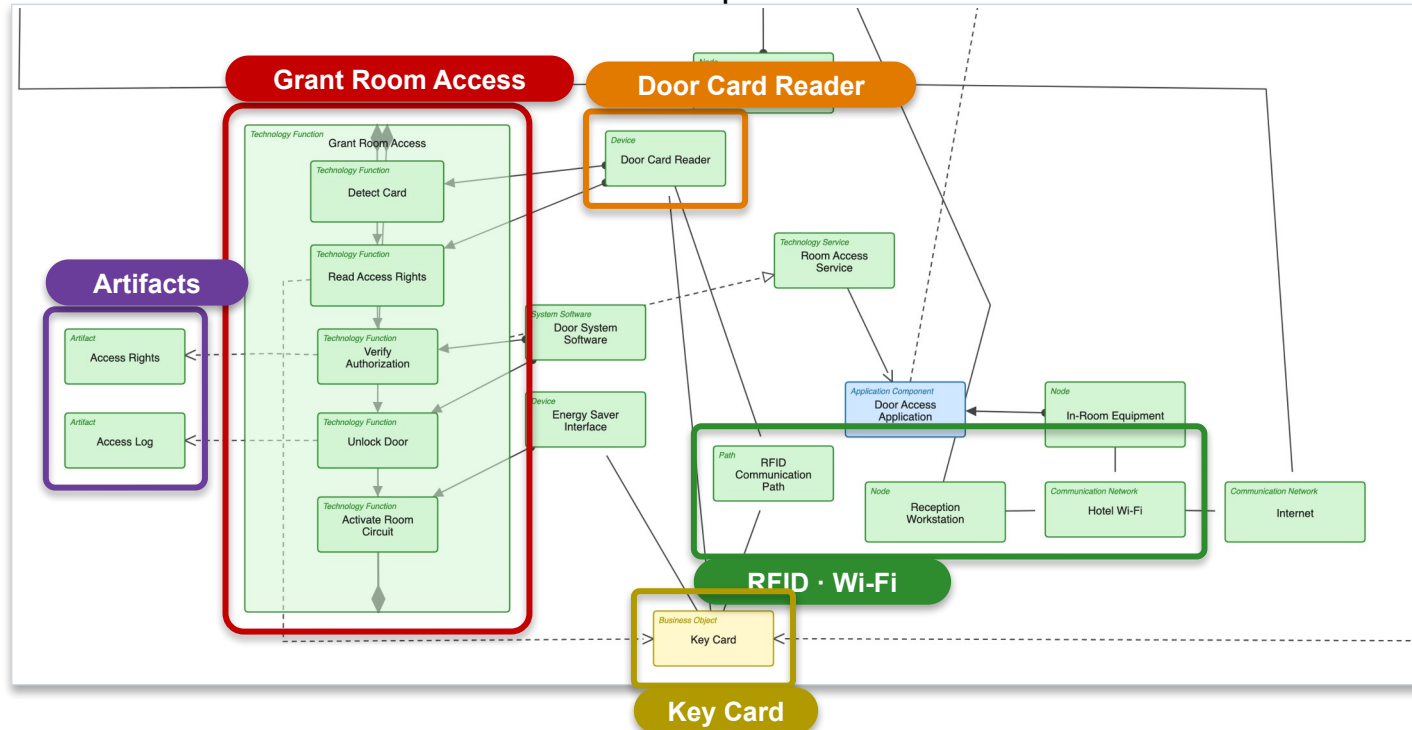
F1 Application Layer — Inside the HMS

The abstract view hid this: functions, services, events and data objects



F1 Technology Layer — Down to the Door

How a reservation made online becomes a door that opens



Value of the EA model

Transparency across 5+ cooperating systems — one shared picture instead of tribal knowledge

Impact analysis becomes concrete: “what breaks if the HMS is down?” — trace the serving arrows

Onboarding: a new receptionist or developer sees where their work sits in the whole

Limitations

A point-in-time snapshot — no processes for keeping it current (governance)

No quantities or SLAs: one AWS box says nothing about capacity or failover

Every abstraction choice hides detail — In-Room Equipment collapses many devices

Reflection — Missing Information & Challenges

Missing in the case material

No infrastructure specs for the reservation side — hosting had to stay abstract

Interfaces to external booking platforms are underspecified

No failure or volume data — capacity questions can't be answered from the model

Our real challenges

Choosing the abstraction level — when is a box one system or five?

Keeping abstract and detailed views consistent while both evolved

Parallel modelling with coArchi: merge conflicts and duplicate elements — we merged 11 duplicates late

Backup

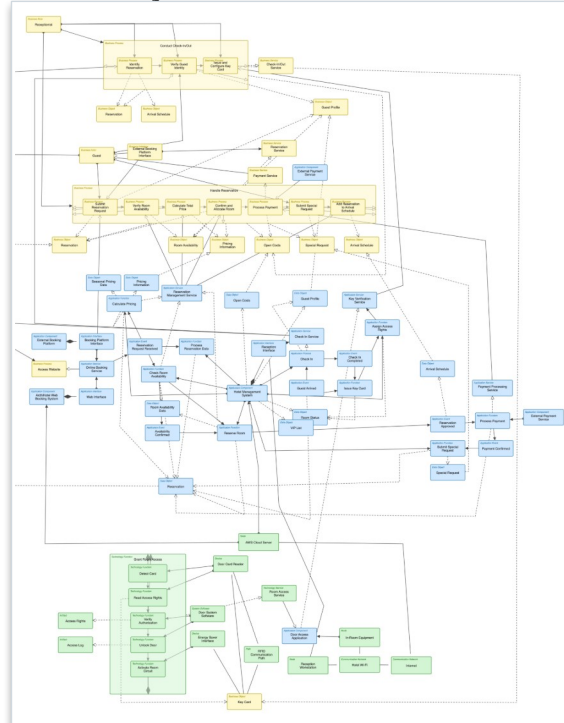
Backup — Detailed View, Complete

Detailed — F1 Reservation & Check-In · 86 objects · 160 connections on the view

All three layers in one view.

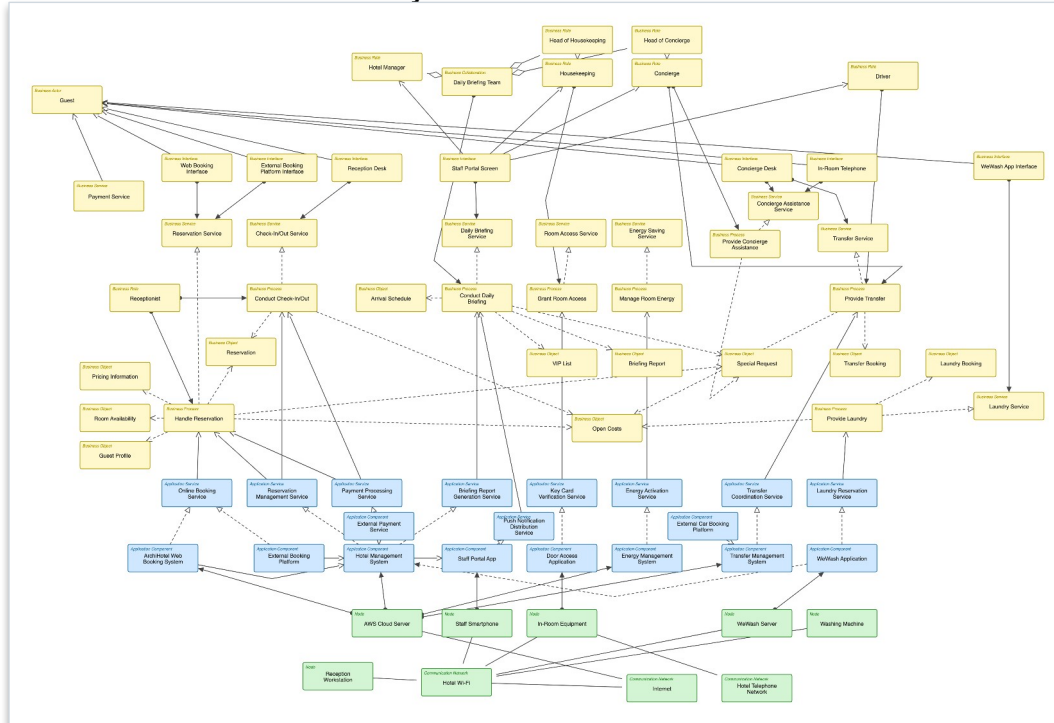
Business on top, application
centre, technology at the
bottom.

PDF export in the submission
shows it at full resolution.



Backup — Abstract View, Complete

Abstract — ArchiHotel EA · all services · 72 objects · 98 connections on the view



The model

One .archimate file · two views (as required)
Abstract view: 72 objects, 98 connections
Detailed F1 view: 86 objects, 160 connections
Naming: noun phrases for structure, verb phrases for behaviour

Tooling & workflow

Archi + coArchi plugin, shared GitHub repository
One layer per person, weekly alignment on the cross-layer relationships
Independent model audit before submission (fix PRs merged)